

PBA: 11. Sampling and Sampling Distributions

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Roadmap

1. Sampling Exercise
2. Sampling Framework and Polls
3. Revisiting Random Variables and Probability Distributions
4. Sampling Distribution
5. Normal Variables and the Central Limit Theorem

1/ Sampling Exercise

Data on Class Years Enrolled in PBA Fall'23

```
1 # don't forget to load tidyverse and dplyr as usual!
2 > class_years <- read_csv("https://bit.ly/47TCVp7"); class_years
3
4 # A tibble: 71 × 5
5   major admission_yr program_yr year is_intl
6   <chr>          <dbl>      <dbl> <chr>      <dbl>
7 1 IPMT          110         3 Junior         0
8 2 TM            112         1 Graduate Year 1 0
9 3 IPMT          109         4 Senior         1
10 4 EECS          109         4 Senior         1
11 5 IPMT          109         4 Senior         0
12 6 TM            112         1 Graduate Year 1 0
13 7 TM            112         1 Graduate Year 1 0
14 8 EECS          110         3 Junior         1
15 9 IPMT          110         3 Junior         1
16 10 ECON         110         3 Junior         0
17 # 61 more rows
18 # Use `print(n = ...)` to see more rows
```

- A total of 70 enrolled students + 1 auditing the course.
 - `is_intl`: indicator for international students.

Composition of the 2023 PBA Roster

Class Year:

```
1 > class_years |>
2   count(year) |>
3   mutate(prop = n / sum(n)) |>
4   arrange(desc(prop))
5
6 # A tibble: 7 × 3
7   year          n prop
8   <chr>      <int> <dbl>
9 1 Senior          24 0.338
10 2 Junior          23 0.324
11 3 Graduate Year 1   16 0.225
12 4 Sophomore         5 0.0704
13 5 Audit Grad. Year 2  1 0.0141
14 6 Graduate Year 2   1 0.0141
15 7 Visiting Grad. Year 2 1 0.0141
```

Major:

```
1 > class_years |>
2   count(major) |>
3   mutate(prop = n / sum(n)) |>
4   arrange(desc(prop))
5
6 # A tibble: 10 × 3
7   major      n prop
8   <chr> <int> <dbl>
9 1 IPMT     33 0.465
10 2 EECS     11 0.155
11 3 TM        9 0.127
12 4 IMBA      8 0.113
13 5 QF        4 0.0563
14 6 ECON      2 0.0282
15 7 CS        1 0.0141
16 8 IBP        1 0.0141
17 9 ISS        1 0.0141
18 10 KPC       1 0.0141
```

Q: What proportion of the class is juniors?

- If we think about the (super)population of all students who could take PBA, what is the true population proportion of juniors?
- We treat the 2023 roster as a finite realization of the population so we can simulate sampling.

Let's Take Some Samples!

- We can use the `slice_sample()` function to take a random sample of rows of a tibble:

Our 1st Sample:

```
1 > class_years |>  
2   select(year) |>  
3   slice_sample(n = 5)  
4  
5 # A tibble: 5 × 1  
6   year  
7   <chr>  
8 1 Sophomore  
9 2 Junior  
10 3 Senior  
11 4 Junior  
12 5 Senior
```

Another Sample:

```
1 > class_years |>  
2   select(year) |>  
3   slice_sample(n = 5)  
4  
5 # A tibble: 5 × 1  
6   year  
7   <chr>  
8 1 Junior  
9 2 Senior  
10 3 Senior  
11 4 Senior  
12 5 Junior
```

- Sample proportion of 3rd year undergrads (n=20):

```
1 > class_years |> slice_sample(n = 20) |>  
2   summarize(jn_prop = mean(year == "Junior"))  
3  
4 # A tibble: 1 × 1  
5   jn_prop  
6   <dbl>  
7 1    0.25
```

Repeated Sampling

- We sometimes want to draw multiple samples from a tibble. For this we can use `rep_slice_sample()` from the `infer` package:

```
1 > library(infer) # don't forget to install the package!
2 > class_years |>
3   rep_slice_sample(n = 5, reps = 2)
4
5 # A tibble: 10 × 6
6 # Groups:   replicate [2]
7   replicate major admission_yr program_yr year is_intl
8   <int> <chr> <dbl> <dbl> <chr> <dbl>
9   1 1 IPMT 110 3 Junior 0
10  2 1 EECS 109 4 Senior 1
11  3 1 IPMT 109 4 Senior 1
12  4 1 IMBA 112 1 Graduate Year 1 1
13  5 1 IPMT 110 3 Junior 1
14  6 2 TM 112 1 Graduate Year 1 0
15  7 2 IPMT 109 4 Senior 1
16  8 2 IPMT 109 4 Senior 0
17  9 2 IPMT 109 4 Senior 1
18 10 2 TM 112 1 Graduate Year 1 0
```

Simulate Many Separate 'Studies' Being Done

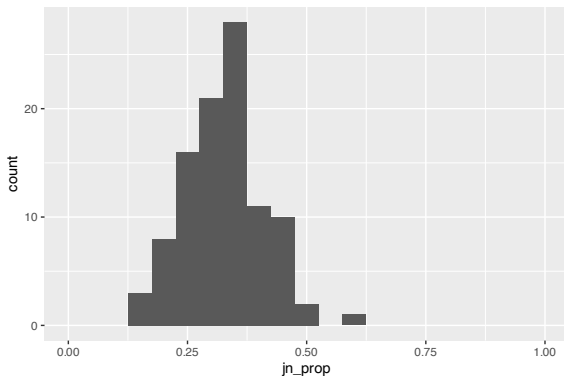
- Each sample/replicate can be thought of as a simulated 'study' of the population

```
1 > samples_n20 <- class_years |>
2   rep_slice_sample(n = 20, reps = 100) |>
3   group_by(replicate) |>
4   summarize(jn_prop = mean(year == "Junior")); samples_n20
5
6 # A tibble: 100 × 2
7   replicate jn_prop
8   <int>     <dbl>
9     1         0.25
10    2         0.3
11    3         0.3
12    4         0.25
13    5         0.25
14    6         0.35
15    7         0.6
16    8         0.3
17    9         0.35
18   10         0.35
19 # 90 more rows
20 # Use `print(n = ...)` to see more rows
```

- Proportion of Juniors: `jn_prop` shows the percentage of Juniors in each sample/replicate, illustrating variability across individual studies.

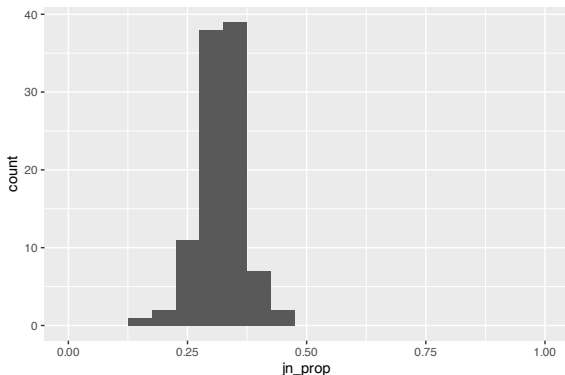
Distribution of These Proportions

```
1 > samples_n20 |>  
2   ggplot(mapping = aes(x = jn_prop)) +  
3   geom_histogram(binwidth=0.05) +  
4   lims(x = c(0, 1))
```



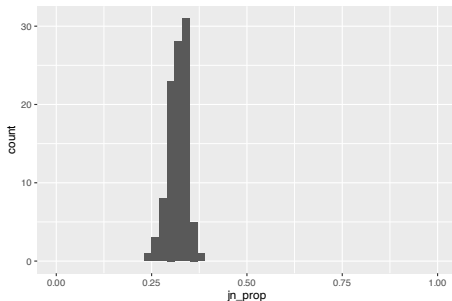
What If the Sample Sizes are Bigger? (n=40)

```
1 > samples_n40 <- class_years |>
2   rep_slice_sample(n = 40, reps = 100) |>
3   group_by(replicate) |>
4   summarize(jn_prop = mean(year == "Junior"))
5
6 > samples_n40 |>
7   ggplot(mapping = aes(x = jn_prop)) +
8   geom_histogram(binwidth=0.05) + lims(x = c(0, 1))
```



What About When n=60?

```
1 > samples_n60 <- class_years |>  
2   rep_slice_sample(n = 60, reps = 100) |> group_by(replicate) |>  
3   summarize(jn_prop = mean(year == "Junior"))  
4 > samples_n60 |>  
5   ggplot(mapping = aes(x = jn_prop)) + geom_histogram(binwidth=0.02) +  
6   lims(x = c(0, 1))
```



- **Sampling distribution** of our estimates (i.e., sample proportions of juniors).
 - $\neq$ **sample's distribution** (of values in a single sample).
 - Q: What would happen when n=70?

Sample Size and Variability Across Samples

- When sample size = 20:

```
1 > samples_n20 |>  
2   summarize(sd(jn_prop)) |> pull()  
3 [1] 0.08241923
```

- Sample size = 40:

```
1 > samples_n40 |>  
2   summarize(sd(jn_prop)) |> pull()  
3 [1] 0.04557711
```

- Sample size = 60:

```
1 > samples_n60 |>  
2   summarize(sd(jn_prop)) |> pull()  
3 [1] 0.02029178
```

↪ The variability of the sample statistic (i.e., `jn_prop`; **sampling variation**) decreases as **sample size** gets larger.

2/ Sampling Framework and Polls

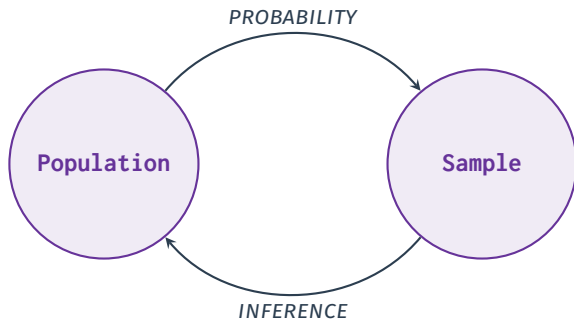
Populations

- **Population:** group of units/people we want to learn about.
- **Population parameter:** some numerical summary of the population we would like to know. - population mean/proportion, population standard deviation.
- **Census:** complete recording of data on the entire population.

Samples

- **Sample:** subset of the population taken in some way (hopefully randomly).
- **Estimator or sample statistic:** numerical summary of the sample that is our “best guess” for the unknown population parameter.

Sampling Framework *Redux*



Sampling at Random

- **Random sample:** units selected into sample from population with a non-zero probability.
 - vs. **non random/probabilistic sampling**, e.g., convenience sampling, volunteer (self-selection) sampling, etc.
- **Simple random sample:** all units have the same probability of being selected into the sample.
 - vs. **stratified sampling** → to assure our samples mirror specific characteristics of the population such as age or income bracket.

Our Sampling Exercise

- **Population:** all students enrolled in PBA Fall 2023.
- **Population parameter:** population proportion of juniors (3rd year undergrad students) enrolled in PBA Fall 2023.
 - Population proportions often denoted p
- **Sample:** simple random sample of different sizes.
- **Sample statistic/estimator:** sample proportion of juniors.
 - Estimators often denoted with a hat: $\hat{p}$
 - We saw the $\hat{p}$ varies with the random sample taken.

Expected Value

The **expected value** of a sample statistic, $\mathbb{E}[\hat{p}]$, is the (long-run) average value of the statistic across repeated samples.

```
1 > samples_n60 |>  
2   summarize(mean(jn_prop)) |> pull() # rep = 100 inducing some diff. from 0.324  
3  
4 [1] 0.327
```

- When we have a simple random sample, the expected value of a sample proportion is equal to the population proportion, $\mathbb{E}[\hat{p}] = p$.
- Caveat: not true when our sample is **biased** in some way!

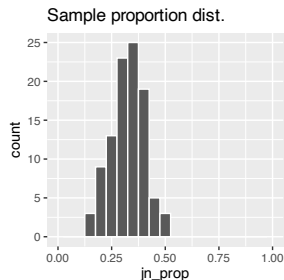
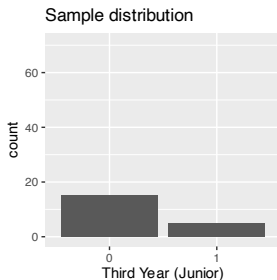
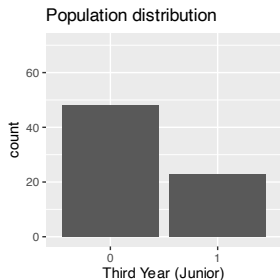
Standard Error Across Repeated Samples

The **standard error** of a sample statistic (e.g., proportion of juniors in class) across repeated samples is the standard deviation of the sample statistic across repeated samples.

```
1 > samples_n60 |>  
2   summarize(sd(jn_prop)) |> pull()  
3  
4 [1] 0.02029178
```

- Tells us how far away, on average, the sample proportion will be from the population proportion.
- **Standard error vs. population/sample standard deviation:**
 - Should not be confused with the population standard deviation or sample standard deviation, both of which **measure how far units are away from a mean!**

The Three Distributions Visualized



Polls: How Popular is Tesla's Cybertruck?



- What proportion of the market is interested in the Tesla Cybertruck?
 - Demand forecasting for a soon-to-be released product
- GLG market research poll/survey:
 - Polling period: late 2019
 - Targeted pickup truck owners (potential customers)
 - Method: e.g., mobile/online survey, telephone interviews (not specified!)
 - 43% are interested in purchasing (vs. 57% not interested)

Poll in Our Framework

- **Population:** potential customers (i.e., pickup truck owners).
- **Population parameter:** Proportion of all potential customers interested in purchasing the Tesla Cybertruck.
 - Census: Not feasible
 - Q: why? $\rightsquigarrow$ (Voluntary) Share your thoughts on Canvas!
- **Sample:** random digit dialing phone numbers (cell and landline).
- **Point estimate:** Sample proportion that is interested in purchasing the Cybertruck.

3/ Revisiting Random Variables and Probability Distributions

Revisiting Random Variables

Random variables are numerical summaries of chance processes.

- *Chance processes*: any procedure/phenomenon that leads to random outcome.

$$X_i = \begin{cases} 1 & \text{if respondent } i \text{ is interested in buying Cybertruck,} \\ 0 & \text{otherwise} \end{cases}$$

- With a **simple random sample**, the **chance of $X = 1$** is equal to the **population proportion** of people that are interested in buying Cybertruck.

Types of Random Variables

- **Discrete:** Can take a finite (or countably infinite) number of values.
 - Number of heads in 5 coin flips.
 - Sampled CEO is a woman ($X = 1$) or not ($X = 0$).
 - Number of battle deaths in a civil war.
- **Continuous:** Can take any real value (usually within an interval).
 - GDP per capita (average income) in a country.
 - Amount of time spent on landing page for a news article.
 - Share of potential customers that are interested in buying Cybertruck.

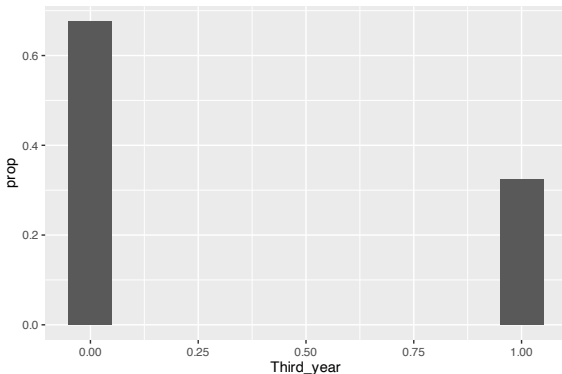
Probability Distributions

- **Probability distributions** tell us the chances of different values of a random variable occurring.
- **Discrete variables:** like a frequency barplot for the population distribution.
- **Continuous variables:** like a continuous version of population histogram.

Discrete Probability Distribution

- We can use the `y = after_stat(prop)` aesthetic to get a barplot with proportions instead of count to show us the chance/probability of selecting a third year undergrad student:

```
1 > class_years |>  
2   mutate(Third_year = as.numeric(year == "Junior")) |>  
3   ggplot(aes(x = Third_year)) +  
4   geom_bar(mapping = aes(y = after_stat(prop)), width = 0.1)
```



Midwest Data *Redux*

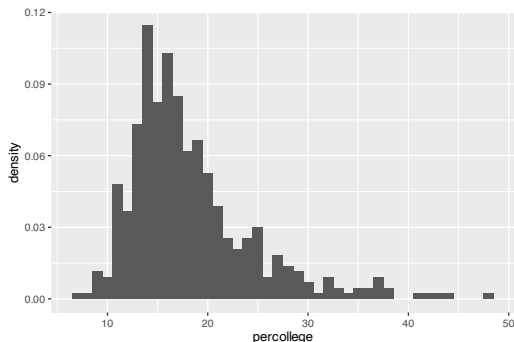
```
1 > library(ggplot2)
2 > midwest
3
4 # A tibble: 437 × 28
5   PID county state area poptotal popdensity popwhite popblack popamerindian popasian popother
6   <int> <chr> <chr> <dbl> <int> <dbl> <int> <int> <int> <int> <int>
7 1 561 ADAMS IL 0.052 66090 1271. 63917 1702 98 249 124
8 2 562 ALEXAN IL 0.014 10626 759 7054 3496 19 48 9
9 3 563 BOND IL 0.022 14991 681. 14477 429 35 16 34
10 4 564 BOONE IL 0.017 30806 1812. 29344 127 46 150 1139
11 5 565 BROWN IL 0.018 5836 324. 5264 547 14 5 6
12 6 566 BUREAU IL 0.05 35688 714. 35157 50 65 195 221
13 7 567 CALHOUN IL 0.017 5322 313. 5298 1 8 15 0
14 8 568 CARROLL IL 0.027 16805 622. 16519 111 30 61 84
15 9 569 CASS IL 0.024 13437 560. 13384 16 8 23 6
16 10 570 CHAMPA IL 0.058 173025 2983. 146506 16559 331 8033 1596
17 # 427 more rows
18 # 17 more variables: percwhite <dbl>, percblack <dbl>, percamerindian <dbl>, percasian <dbl>,
19 # percother <dbl>, popadults <int>, perchsd <dbl>, percollege <dbl>, percprof <dbl>,
20 # poppovertyknown <int>, percpovertyknown <dbl>, percbelowpoverty <dbl>,
21 # perchildbelowpovert <dbl>, percadultpoverty <dbl>, percelderlypoverty <dbl>, inmetro <int>,
22 # category <chr>
23 # Use `print(n = ...)` to see more rows
```

- Data on socio-demographic info. for midwest counties from 2000 US census.
- percollge: Percent of population in a given county that are college educated.

Continuous Probability Distribution

- We can use the `y = after_stat(density)` to create a **density histogram** instead of a count histogram.
 - ↪ The area of the histogram boxes are equal to the chance of randomly selecting a unit in that bin:

```
1 > midwest |>  
2   ggplot(aes(x = percollege)) +  
3   geom_histogram(aes(y = after_stat(density)), binwidth = 1)
```



Why Use Density?

- Histograms with **density** on the y-axis are drawn in a way that the area of each box is equal to **the proportion of units in the sample** in that horizontal bin.
- Easier to compare distributions across different sample sizes.
- Sum up all the area = 1 (but heights can go above 1)
 - E.g., bin width below 1 w/ large number of data points.

4/ Sampling Distribution

Key Properties of Sums and Means

- Suppose $X_1, X_2, \dots, X_n$ is a simple random sample from a population distribution with mean μ (“mu”) and variance σ^2 (“sigma squared”)
 - e.g., X_i is share of residents in county i with college education.

- **Sample mean:** $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

...

- $\bar{X}_n$ is a random variable with a distribution!!

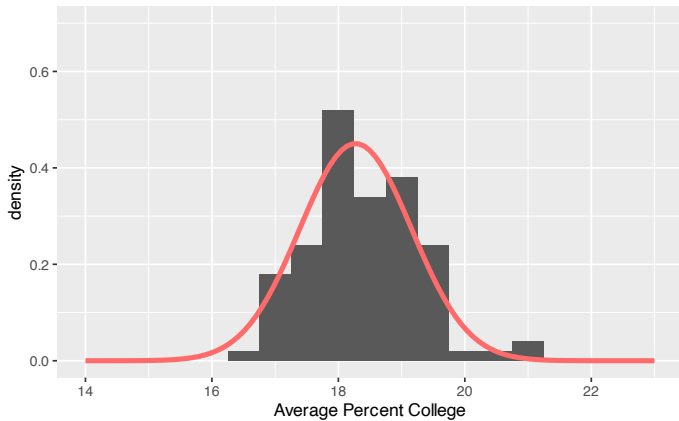
Sample Means/Proportions Distribution

Sampling distributions are the probability distributions of an estimator like $\bar{X}_n$

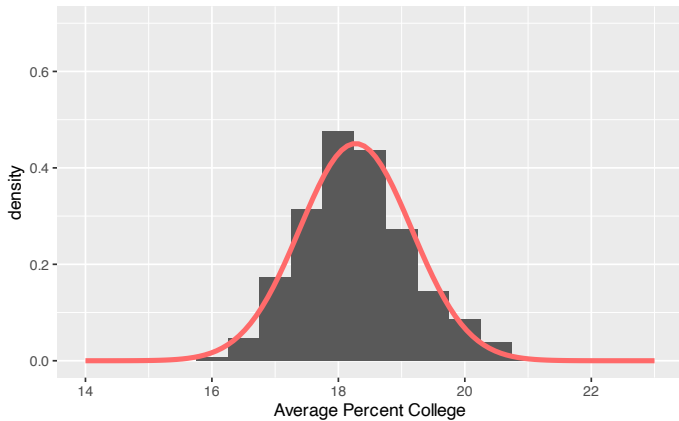
- When we have access to the full population, we can approximate the sampling distribution with repeated sampling.

```
1 > midwest |>
2   rep_sample_n(size = 50, reps = 100) |>
3   group_by(replicate) |>
4   summarize(`Average Percent College` = mean(percollege)) |>
5   ggplot(mapping = aes(x = `Average Percent College`)) +
6   geom_histogram(aes(y = after_stat(density)), binwidth = 0.5) +
7   coord_cartesian(xlim = c(14, 23), ylim = c(0, 0.7)) +
8   labs(title = "100 Repetitions") +
9   stat_function(fun = dnorm, args = c(mean = mean(midwest$percollege),
10                                         sd = sd(midwest$percollege)/sqrt(n)),
11               color = "indianred1", size = 1.5, xlim = c(14,23))
```

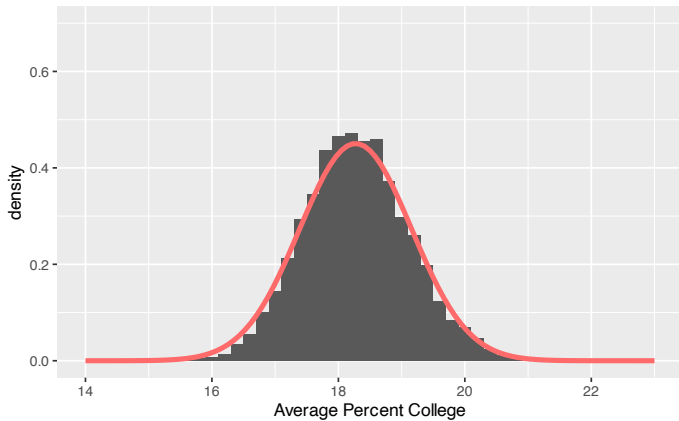
100 Repetitions

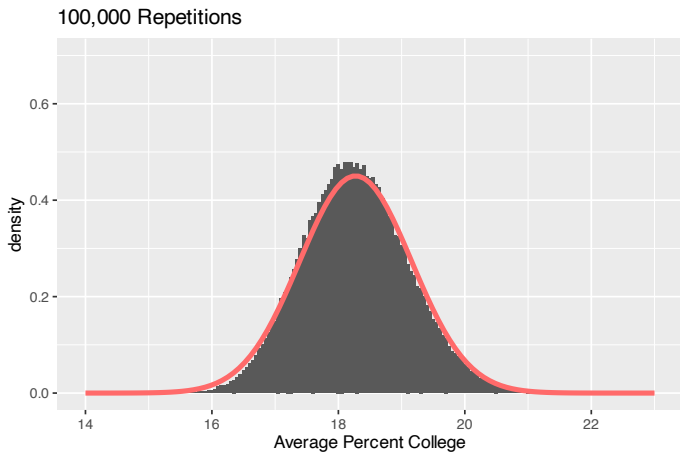


1,000 Repetitions



10,000 Repetitions





Better approximation of the sampling distribution with more repetitions.

Sampling Distribution of the Sample Mean

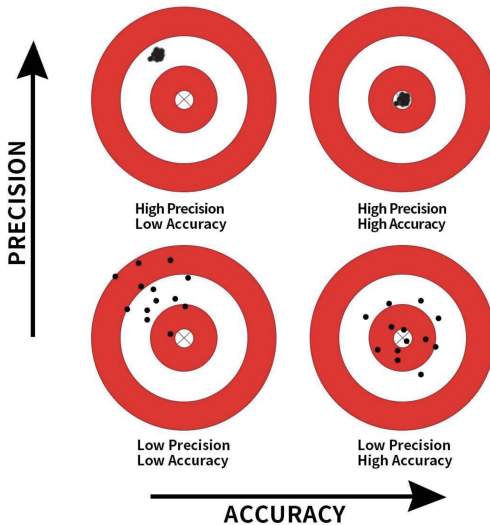
- Suppose $X_1, X_2, \dots, X_n$ is a simple random sample from a population with mean μ and finite variance σ^2 .
- **Expected value** of the distribution of $\bar{X}_n$ is the population mean, μ .
- **Standard error** of the distribution of $\bar{X}_n$ is approximately $\sigma/\sqrt{n}$:

$$SE \approx \frac{\text{population standard deviation}}{\sqrt{\text{sample size}}}$$

Unbiasedness

- An estimator is **unbiased** when its expected value across repeated samples equals the population parameter of interest.
- Sample mean of a **simple random sample** is **unbiased** for the population mean, $\mathbb{E}[\bar{X}_n] = \mu$.
- An estimator that isn't unbiased is called **biased** and thus is not representative of the population parameter.

Precision vs. Accuracy



Source: <https://www.forecast.app/blog/difference-between-accuracy-precision>

Law of Large Numbers

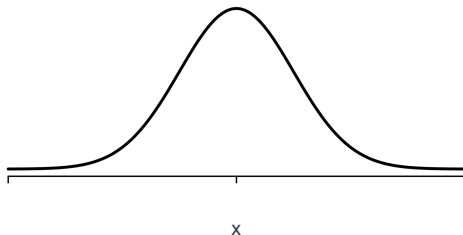
Law of Large Numbers (LLN)

Let $X_1, \dots, X_n$ be a simple random sample from a population with mean μ and finite variance σ^2 . Then, $\overline{X}_n$ converges to μ as n gets large.

- Probability of $\overline{X}_n$ being “far away” from μ goes to 0 as n gets big.
- The distribution of sample mean “collapses” to population mean.
- Can see this from the SE of $\overline{X}_n$: $SE = \sigma/\sqrt{n}$.
- Not necessarily true with a biased sample!

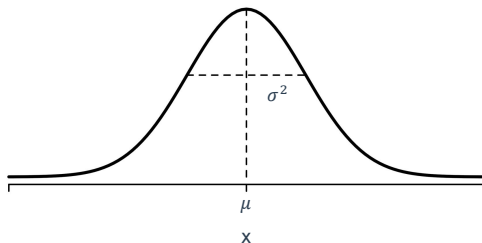
5/ Normal Variables and the Central Limit Theorem

Normal Random Variable



- A **normal distribution** has a PDF that is the classic “bell-shaped” curve.
 - Extremely ubiquitous in statistics.
 - An r.v. is more likely to be in the center, rather than the tails.
- Three key properties of this PDF:
 - **Unimodal**: one peak at the mean.
 - **Symmetric** around the mean.
 - **Everywhere positive**: any real value can possibly occur.

Normal Distribution



- A normal distribution can be affected by two values:
 - **mean/expected value** usually written as μ
 - **variance** written as σ^2 (standard deviation is σ)
 - Written $X \sim N(\mu, \sigma^2)$.
- **Standard normal distribution:** mean 0 and standard deviation 1.

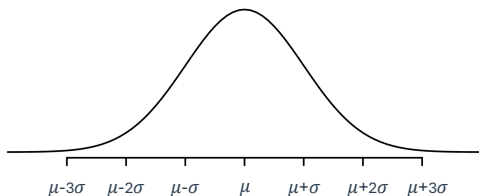
Central Limit Theorem

Central Limit Theorem (CLT)

Let $X_1, \dots, X_n$ be a simple random sample from a population with mean μ and variance σ^2 . Then, $\bar{X}_n$ will be approximately distributed $N(\mu, \sigma^2/n)$ in large samples.

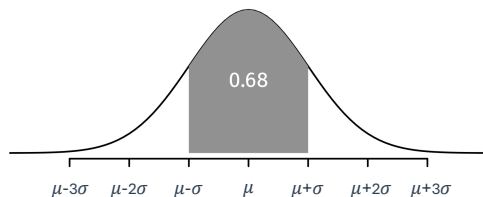
- “Sample means tend to be normally distributed as samples get large.”
- $\rightsquigarrow$ we know (an approx. of) the entire probability distribution of $\bar{X}_n$
 - Approximation is better as n goes up.
 - Does not depend on the distribution of X_i !

Empirical Rule for the Normal Distribution



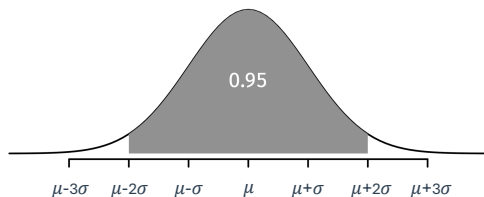
- If $X \sim N(\mu, \sigma^2)$, then:

Empirical Rule for the Normal Distribution



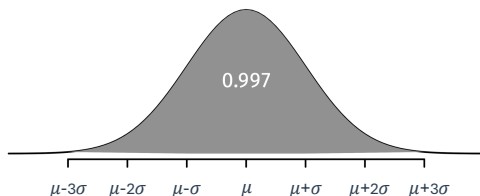
- If $X \sim N(\mu, \sigma^2)$, then:
 - $\approx 68\%$ of the distribution of X is within 1 SD of the mean.

Empirical Rule for the Normal Distribution



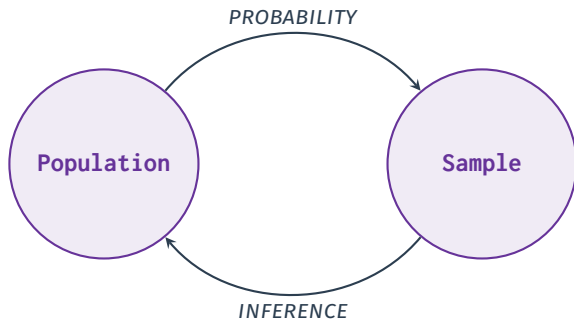
- If $X \sim N(\mu, \sigma^2)$, then:
 - $\approx 68\%$ of the distribution of X is within 1 SD of the mean.
 - $\approx 95\%$ of the distribution of X is within 2 SDs of the mean.

Empirical Rule for the Normal Distribution



- If $X \sim N(\mu, \sigma^2)$, then:
 - $\approx 68\%$ of the distribution of X is within 1 SD of the mean.
 - $\approx 95\%$ of the distribution of X is within 2 SDs of the mean.
 - $\approx 99.7\%$ of the distribution of X is within 3 SDs of the mean.
- CLT + empirical rule: we'll know the rough distribution of estimation errors we should expect.

Where Are We Going?



- We only have one sample to work with.
 - Q: Can we learn about the population from that sample?

Enjoy Your Weekends! :)

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<https://j1yoo4.github.io/>

